Steve Gottlieb
Going Deep

Digging Deep in Messier 83

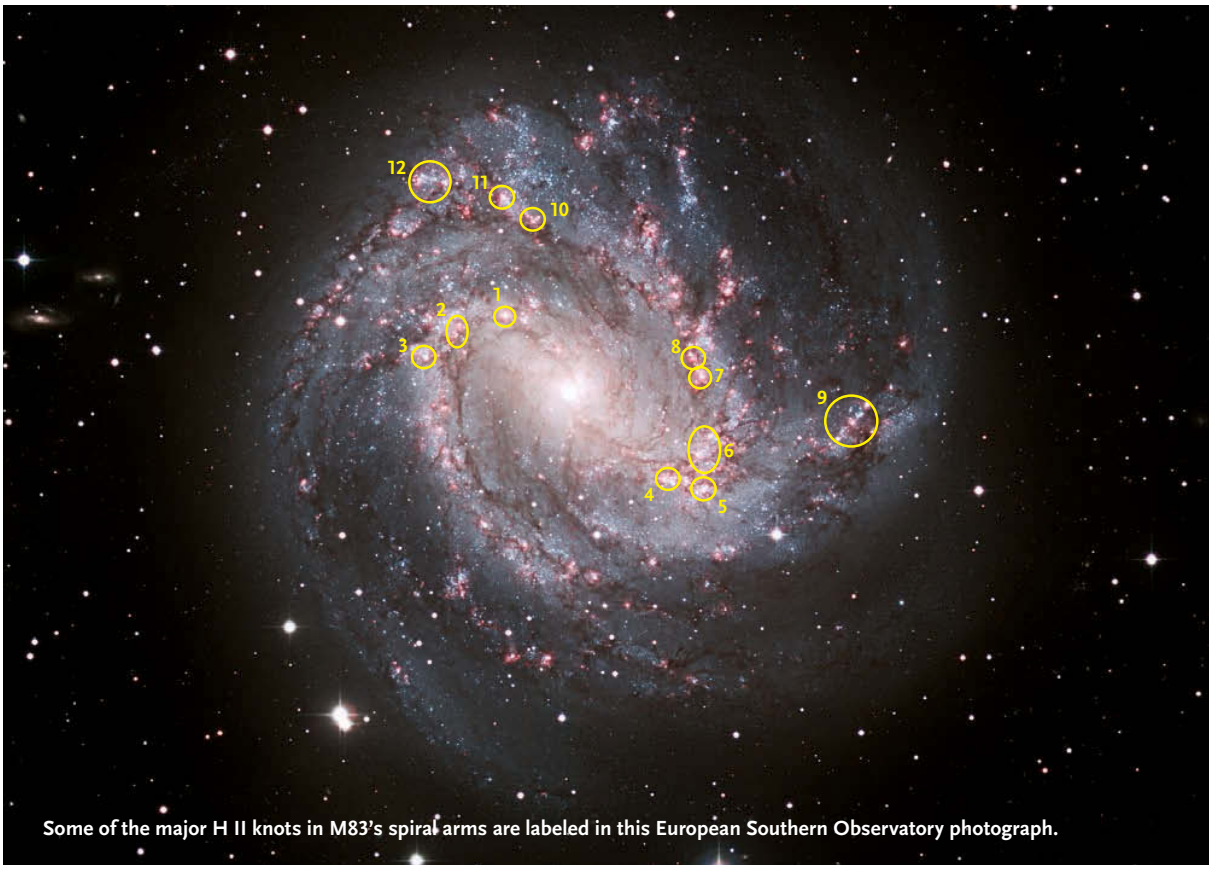
This magnificent spiral galaxy is best observed from southerly latitudes.

M83, THE GORGEOUS Southern Pinwheel, ranks high on my all-time favorites list. Shining at magnitude 7.5, M83 is the 4th-brightest Messier galaxy after M31, M33, and M81. And at a distance of 15 million light-years, it's one of the nearest "grand-design" barred spirals.

M83's spiral arms are bursting with pink star-forming regions and recently minted blue star clusters. A 1983 catalog in the *Astrophysical Journal* lists 298 H II regions — sites where young, massive stars have ionized nearby gas clouds. M83's central bar funnels gas into a starburst nucleus, which harbors numerous massive stars. This activity has produced six supernovae in the past 90 years, a record only surpassed by NGC 6946.

French astronomer Nicolas-Louis de Lacaille discovered M83 during his expedition to the Cape of Good Hope in 1751–1752. Using just a half-inch refractor at 8×, Lacaille cataloged 9,766 southern stars along with 42 nonstellar objects (mostly clusters). He logged M83 as a "small, shapeless nebula" and included it in his list of nebulae "not accompanied by any star."

In 1787 William Herschel trained his 18.7-inch speculum-metal reflector on M83 from his garden in Slough, England. Although the galaxy was just 10° above the horizon, he noted "faint branches about 5′ or 6′ long, elongated southwest to northeast." Searching for clear, transparent skies, British amateur astronomer William



Some of the major H II knots in M83's spiral arms are labeled in this European Southern Observatory photograph.

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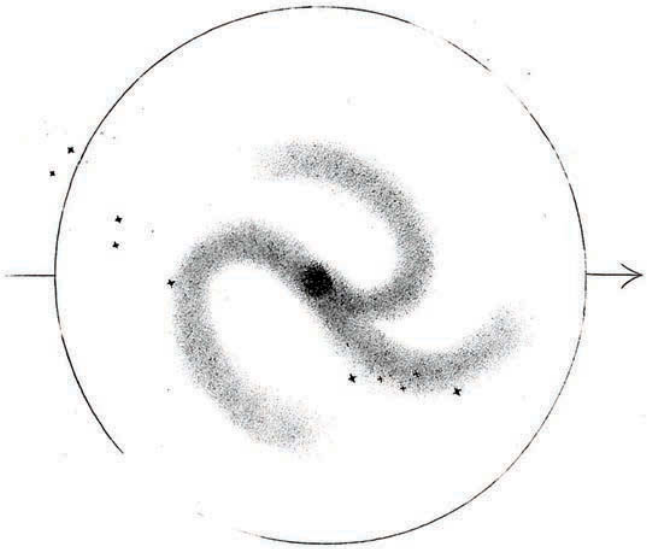


A 48-inch reflector and an observing site at latitude 35° north enabled William Lassell to discern M83's spiral structure in 1862.

Lassell shipped his 48-inch equatorial reflector to the island of Malta in 1861. Lassell observed M83 the following spring, describing it as a "three-branched spiral" and producing the sketch shown above.

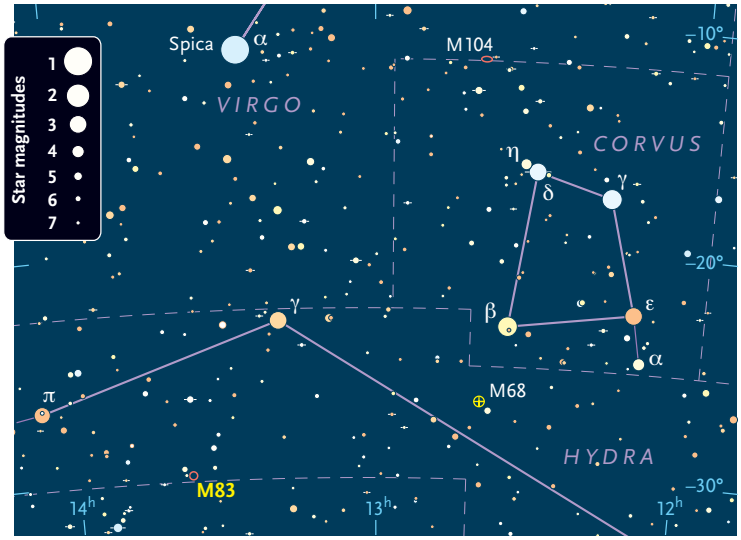
M83 is buried in southern Hydra, 19° south of Spica, at declination –30°. So mid-northerners often observe M83 through the haze and light pollution that tend to hug the horizon, making it tough to see more than its bright core. But given the right conditions, M83 is rich in detail, with 8- to 10-inch scopes revealing the central bar and spiral arms, and larger apertures resolving multiple H II knots and dark rifts.

I first observed M83's spiral structure through my 13-inch solid-tube reflector in the early 1980s. I saw a weak central bar running northeast to southwest, punctuated by a small, bright core. The two principal arms weren't easy, but they nearly merged to form the Greek letter theta (θ). Not bad from Northern California, but I knew I was missing out on the real deal.

In February 2004, I lugged a 13-inch airline-portable travelscope to the Costa Rica Southern Sky Party, which is held annually at a dark site along the Gulf of Nicoya. From a latitude of +10° north, M83 rose 50° above the horizon, and the detail now popped! Three spiral arms were separated by darker ribbons of dust and embedded within an 8′ × 6′ halo. The arm attached to the northeast side of the bar unfurls counterclockwise around the south side of the galaxy. Its counterpart begins on the bar's southwest end and shoots north before curling eastward along the north side. The third arm branches outside to the south and curves west.

In 2008 I traveled to Zane Hammond's Magellan Observatory in Australia and viewed M83 in his 24-inch f/3.7 scope while it was directly overhead. At 200× the bar extended 3′ × 1′ and enclosed a bright circular nucleus. The two prominent arms appeared chiseled with high contrast by dust lanes at their edges. The longer northern arm winds roughly 270°, merging into the halo on the southeast side near the 8″-wide double star h4599, whose components shine at magnitude 10.5 and 11.5. The southern arm attaches at the northeast side of the bar, sweeps south and west, and peters out 3′ south of center. When I increased the magnification to 375×, several small knots sparkled near the inner ends of both arms, and darker veins of dust sliced across the arms.

Last spring I scrutinized M83 in Texas through Jimi Lowrey's 48-inch jumbo-sized Dob, and the arms were chock-full of glowing H II regions. The brightest of these are marked on the photograph on the facing page. At 488× I saw three prominent knots (#1–3), varying in size from 10″ to 20″, at the northeast end of the central bar. A series of fuzzy splotches was symmetrically placed on the opposite side (#4–8), with the largest splotch extending 25″. And I saw additional bright clumps at the southwest (#9) and northeast (#10–12) sides of the halo.

To view M83 in its full glory — along with the many other showpieces of the far-southern sky — I highly recommend adding a Southern Hemisphere trip to your bucket list. But from any location, your best view will be on a dark, transparent night when M83 is at its highest on the meridian. ✦

S&T *contributing editor* ***Steve Gottlieb*** *has made several trips to the Southern Hemisphere to complete his observations of the* New General Catalogue of Nebulae and Star Clusters.